Lenin T

Coimbatore, Tamil Nadu

+91 8220110903 | leninrithick@gmail.com | linkedin.com/in/lenin-t/ | github.com/Lenin-afk | Portfolio

Summary

ECE graduate with hands-on project experience in Python development, backend engineering, REST APIs, database integration, and machine learning. Built applications using Python, FastAPI, SQL, and Scikit-learn. Seeking entry-level Software Development Engineer or AI/ML Engineer roles to build practical solutions to real-world problems.

Education

Kumaraguru College of Technology **Nov 2022 – May 2026**

B.E. in Electronics and Communication Engineering – CGPA: 7.90 *Coimbatore, Tamil Nadu*

Sri Vivekananda Matriculation Higher Secondary School

Class XII – 86.6% — *Class X – 96.2%* *Coimbatore, Tamil Nadu*

Technical Skills

Languages & Core Concepts: Python; SQL; C; Object-Oriented Programming; Data Structures and Algorithms

Software & Web Development: FastAPI; REST APIs; SQLAlchemy; JWT Authentication; React; HTML; CSS;

Databases & Data Analysis: PostgreSQL; SQLite; NumPy; Pandas; Matplotlib; Data Preprocessing; Exploratory Data Analysis

Machine Learning & Generative AI: Scikit-learn; Random Forest; Model Evaluation; Classification Metrics; PyTorch; CNN Fundamentals; Generative AI Fundamentals; LLM Fundamentals; Prompt Engineering

Tools & Technologies: Selenium; Git; GitHub; Docker; Linux; VS Code

Projects

PredictIQ – Predictive Maintenance System **Oct 2026**

Tech Stack: Python, Scikit-learn, Pandas, NumPy, FastAPI, SQLite, Docker, Render

- Built a Python-based predictive maintenance application using a Random Forest model and the AI4I 2020 dataset to predict potential machine failures.
- Used FastAPI to expose model predictions and SQLite to store prediction history.
- Evaluated model performance and achieved 98.15% test accuracy on the dataset.
- Containerized the application with Docker.

ToLet Coimbatore – Hyperlocal Rental Platform **Apr 2026 – Jun 2026**

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, JWT Authentication, Cloudinary, Railway

- Developed a hyperlocal rental listing platform for property listings and tenant-landlord interactions in Coimbatore.
- Implemented REST APIs and database operations for property listing CRUD and user authentication using FastAPI, SQLAlchemy, PostgreSQL, and JWT.
- Integrated Cloudinary for property photo uploads and deployed the application on Railway using environment variables and a cloud PostgreSQL database.

Vehicle Platooning for Heavy Duty Vehicles **Dec 2025 – Apr 2026**

Tech Stack: ESP32, Arduino UNO, ESP-NOW, Ultrasonic Sensor, L298N Motor Driver

- Developed a leader-follower vehicle communication prototype using ESP-NOW for real-time coordination.
- Integrated ultrasonic-based obstacle detection with an automatic braking mechanism.
- Demonstrated synchronized vehicle movement for improved transportation efficiency and safety.

Certifications

- Introduction to Artificial Intelligence (AI) – IBM, Coursera
- Generative AI: Introduction and Applications – IBM, Coursera
- Introduction to Internet of Things – NPTEL (IIT Kharagpur), Elite (91%)